The Phonology of Indonesian

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Key Points

- Description of the segmental inventory; restrictions on segments; phonology of /ʔ h ə /
- Discussion of nasal substitution (Pater, 1999); connection to parallel research on Malay
- Discussion of word-level prominence; evidence for metrical structure; prosodic phrasing

Abstract

This paper reviews the phonology of Standard Indonesian, focusing on the following set of topics: (*i*) the segmental inventory, (*ii*) the special segments /ʔ h ə/, (*iii*) restrictions on syllable structure, (*iv*) positional constraints on segments, (*v*) nasal substitution, (*vi*) reduplication, (*vii*) the status of word-level prominence, (*viii*) foot structure, and (*ix*) higher-level prosodic domains. The discusison weighs the facts of the standard language against two members of the local linguistic ecosystem, Javanese and Jakarta Indonesian, and concludes with a reflection on the emerging direction of work on word-level stress.

Introduction

Bahasa Indonesia, or *Indonesian*, is the national language of Indonesia. It belongs to the Malayic subgroup of Malayo-Polynesian, a higher-order subgroup of Austronesian, and it is closely related to the varieties of Malay official in Malaysia, Singapore, and Brunei. The language originates most directly from the literary Malay of the courts of Malacca, Johore, and Riau, and it was cultivated as the main language of administration across modern-day Indonesia by the Dutch and then cemented as the primary vehicle of the nationalist movement in 1928 (Adelaar, 2000). The Indonesian Bureau of Statistics reports that it is now spoken natively by around 71,000,000 people and as a second language by 176,500,000 others (Badan Pusat Statistics Indonesia, 2020).

Standard Indonesian is an acrolectal language that must be studied and discussed against a complex sociolinguistic backdrop. There are over 700 regional languages spoken in Indonesia today and the standard language interacts in an especially tight fashion with substrates of four types: (*a*) regional lingua francas that are lexified by Indonesian but borrow from local languages across all systems of the grammar, (*b*) Jakarta Indonesian, the lingua franca that has emerged around the capital (Wallace, 1976), (*c*) Javanese, the regional language with the greatest number of native speakers, highest prestige, and widest distribution across the archipelago, and (*d*) the Javanese varieties of Indonesian that are prominent across all domains of media, government, and public life. Speakers of Standard Indonesian typically have command over substrates of this type, in addition to any regional languages they may already speak, and even those who are not native speakers of Javanese and Jakarta Indonesian are often well-exposed to both. For this

reason, speakers are likely to draw on different and varying linguistic subsystems when asked to produce and provide judgments about the acrolectal standard, and as a result, there is disagreement in the literature over many properties of the language. The goal of this review, then, is to organize and present the generalizations most widely accepted of the standard language, highlight the central domains where descriptions diverge, and frame the analytical challenges that persist against the facts of Jakarta Indonesian and Javanese.

Consonants

Table (1) presents the consonantal phoneme inventory of Standard Indonesian, using parentheses for the phonemes that appear only in the loan strata of the lexicon. The language contrasts three series of stops and a series of palatal affricates, along with four nasals at the corresponding points of articulation; as in many languages of Western and Central Indonesia, the segments /*t n*/ are dental and the segments /*d r*/ are postalveolar (Blust, 2013). The loan strata of the language contain several phonemes not found in the native vocabulary, including the glides /*w j*/ (Adelaar, 1985) and the four fricatives /*f ɕ z x*/. In the earliest loan strata, these fricatives were frequently replaced with native equivalents, and as a result, the standard language contains many doublets that show the alternations *f-p* (*hafal/hapal* "memorize"), *ɕ-s* (*syiwa/siwa* "Shiva"), *z-d͡z/s* (*jaman/zaman* "era"), and *x-k/h* (*khawatir/kuatir* "worry"). Even so, it is clear that many of these segments are now phonemic (pace Alieva et al. (1972)): in more recent loan strata, for instance, there are many minimal pairs for *p/f* (*fakta* 'fact' vs. *pakta* 'pact') and English loans are increasingly adapted with unassimilated /*z ɕ* (< ʃ)/ (Mahdi, 1981; Stokhof, 1975).

(1) *Table One: Consonant Phonemes of Standard Indonesian*

	Bilabial		*Labiodental*	*Dental*		*Postalveolar*	*Palatal*	*Velar*		*Glottal*
Stop	p	b		t		d		k	g	ʔ
Affricate							t͡ɕ d͡z			
Nasal	m			n			ɲ	ŋ		
Fricative			(f)	s	(z)		(ɕ)	(x)		h
Liquid				l						
Rhotic						r				
Glide	w						j			

Glottals

Standard Indonesian has two glottals, /h ʔ/. The segment /h/ is unquestionably phonemic and surfaces in onset positions, in word-final coda positions (e.g., *basah* 'wet'), and in word-medial coda positions at morpheme junctures in the native stratum (e.g., basah-kan 'make wet') and morpheme-internally in many borrowings from Arabic (e.g., *tahta* 'throne'). Lapoliwa (1981) describes several patterns of /h/-deletion that seem, to my judgment, to be sensitive to the positions of words in higher prosodic domains: word-final /h/ is often deleted; root-final /h/ deleted before suffixes (/*kə-lətih-an*/ 'tiredness' > [*kələtian*]) unless the result would be hiatus of identical vowels (/*kə-basah-an*/ 'wetness' > [*kəbasa**h**an*]), and intervocalic /h/ deleted between high-low and low-high sequences (/*lihat*/ 'see' > [*liat*]; /*d͡zahit*/ 'sew' > [*d͡zait*]; cf. /*sehat*/ 'healthy' > [*sehat*]) unless the root-final segment is a liquid (/*lahir*/ 'born' > [*lahir*]). Lapoliwa (1981) claims further that the process is category-sensitive and fails to occur in nouns (thus /*tahu*/ 'know' (V) > [*tau*] but /*tahu*/ 'tofu' (N) > [*tahu*]).

The status of /ʔ/ is more complex. In the Arabic loan stratum, there are a number of roots that must contain underlying instances of /ʔ/, often within word-medial clusters (e.g., jumʔ at 'Friday'). In word-final position, the language also has several minimal pairs that straddle different strata and contrast /ʔ/ with /k/: for instance, /paʔ/ 'father' - /pak/ 'pack' and /masaʔ/ 'no way' - /masak/ 'cook' (and see also (Tadmor, 2007, pp. 304–305). These facts suggest that the segment must be considered phonemic (Mahdi, 1981; Stokhof, 1975; Lapoliwa, 1981). Even so, there are many patterns to suggest that the synchronic phonology contains several regular processes of glottal epenthesis. At the left edge of the prosodic word, to begin, glottal stops consistently appear before vowel-initial roots: preventing resyllabification in compounds (thus /*tanah-air*/ 'motherland' > [*tana ʔair*]: (Mahdi, 1981) and occurring, in careful speech, between verbal roots and certain prefixes (thus /*bər-ɛfɔlusi*/ 'evolve' > [bərʔɛfɔlusi]; /*di-ambil*/ 'taken' > [*diʔambil*]: Cohn & McCarthy, 1998). In borrowings from Arabic, in turn, glottal stops often appear between sequences of like vowels (*sāʔāt* 'time') and mid-low sequences (*soʔal* 'matter'), and in the native stratum, they are consistently inserted to break up identical vowel sequences before the nominalizing suffix *-an* (thus /*ba͡tɕa-an*/ 'reading' > [ba͡tɕaʔan] but /*tari-an*/ 'dance' > [tarijan]). Like /h/, it is liable to deletion in intervocalic positions in a manner that seems sensitive to prosodic phrasing.

Vowels

Table (2) presents the vowel inventory of the language, using parentheses for the phonemes that appear only in the non-native strata. In the native stratum, Standard Indonesian contrasts six vowels: /*i ɛ a ɔ u ə*/ (Alieva et al., 1972; Halim, 1974; Lapoliwa, 1981). The mid-open vowels /ɛ ɔ/ contrast with /*i u*/, though historically they did not (Adelaar, 1985), and the language contains many doublets that show alternations between the two (*senin/senen* 'monday', *lubang/lobang* 'hole'). In the synchronic phonology, Lapoliwa (1981) claims that the mid-open vowels are underlyingly mid-close, represented as /*e o*/, and proposes that they lower to /ɛ ɔ/ in closed final syllables. There is good reason to believe, however, that there are underlying contrasts between mid-open and mid-close vowels in at least the loan strata of the language. In Javanese, to begin, there is an underlying contrast between mid-open and mid-close vowels that is faithfully preserved in borrowings into many forms of Indonesian, and in the standard language, the contrast between them is treated as phonemic in many works by native speakers of Javanese, like Samsuri (1960) and Dardjowidjojo (1966). In loanwords from English and Dutch, the same contrasts are also typically preserved, and in the corresponding strata of the lexicon, Mahdi (1981, pp. 414) adduces a series of minimal pairs: for instance /*kopi*/ 'copy' –/*kɔpi*/ '*coffee*.' In the same stratum, Mahdi (1981) also notes that there is a contrast between /*i*/ and /*ɪ*/, citing the minimal pairs /*d͡ʒɪp*/ 'jib' – /*d͡ʒip*/ 'jeep' and /*nasi tɪm*/ 'rice broth' – *tim* 'team.'

(2) *Table Two: Vowel Phonemes of Standard Indonesian*

	Front		*Mid*	*Back*
High	i	(ɪ)		u
Mid-Close	(e)			(o)
Mid-Open	ɛ		ə	ɔ
Low			a	

Schwa

There are a number of prosodic regularities that surround the distribution of schwa. In the native vocabulary, to begin, it is banned from in word-final syllables, though this constraint is not enforced in loans from Javanese and Dutch (e.g., /haltə bəs/ 'bus stop'). In the same vein, it is the only vowel that surfaces in antepenultimate positions in morphologically simple minimal words in the native vocabulary (Adelaar, 1985). In the synchronic phonology of that stratum, as a result, Lapoliwa, 1981 proposes a regular process of antepenultimate reduction to schwa. Grijns (1981) and Cohn (1989) also propose several patterns of schwa-epenthesis to capture generalizations about its distribution in loanwords from English and Dutch, while Adisasmito (1993), shows that the same vowel must also be posited as an underlying phoneme to capture its distribution in at least some native and borrowed roots.

Nasalization

Nasalization is not contrastive in Standard Indonesian, but the language shows two patterns of nasalization of note. First, nasal consonants trigger nasalization of the immediately following vowel, in a manner that extends farther in related languages (Court, 1970; for formal discussion, Walker, 1998; McCarthy, 2004). In the same vein, nasalization has been claimed to emerge between sequences of low vowels separated by the glottal stop in loanwords from Arabic (Court, 1970; Lapoliwa, 1981): thus *sāʔāt* 'time,' *tāʔãt* 'faithful.'

Syllable Structure

The maximally articulated syllable structure in the native vocabulary was historically (C)V(C); within the various loan strata it now reaches (C)(C)(C)V(C)(C). The language historically restricted CC clusters to word-medial positions, and there it allowed only *r*C, NC (where N = homorganic nasal), and /*ŋs*/. In the native vocabulary and the earliest strata of borrowings, all other medial clusters were reduced: thus Arabic *mumkin* 'possible' > *muŋkin*. In more recent strata of borrowings, however, the language has begun to retain clusters of many more types. Word-medially, for instance, it now allows heterorganic sequences of (non-palatal) nasal + stop: thus *jamrud* 'emerald,' *tanpa* 'without,' *angpao* 'gift envelope'. In onset positions, in the same vein, the loan strata from English, Dutch, and Javanese allow clusters of stop and liquid (*kredit* 'credit,' *pretel* 'dismantle,') and in loans from English and Dutch, the language also allows clusters of sibilant and stop (*stasiun* 'station,' *skala* 'scale') and sibilant, stop, and liquid (*skripsi* 'undergraduate thesis'). In word-final position, stop-sibilant clusters were reduced in early loans from English and Dutch (thus English *sport* > *spor*), but now they are increasingly retained (thus English *complex* > *kompleks*, *index* > *indeks*). In the same positions, liquid-nasal clusters were

once often split by /ɛ/-epenthesis (*modern* > *mɔdɛrn* ~ *mɔdɛrɛn*), but the trend is now toward unassimilated retention (Adisasmito, 1993).

Positional Restrictions on Segments

Standard Indonesian prohibits syllable-final /t͡ɕ d͡z/ and only seems to allow syllable-final /ɲ/ in the word-medial sequences /ɲt͡ɕ/ and /ɲd͡z/ (e.g., *aɲt͡ɕaŋ* 'threaten'). If word-medial homorganic NC-sequences are parsed as complex onsets, then (as they are in Javanese), the syllable-final position may be said to impose a total ban on all palatal segments. The word-final position is traditionally held to ban voiced obstruents as well (Lapoliwa, 1981), in light of the fact that voicing alternations operate in that position in certain strata of loans (thus /*d͡zawab*/ 'respond' > [*d͡zawap*] but /*d͡zawab-an*/ 'response' > [*d͡zawaban*]). In recent borrowings from English and Javanese, however, this prohibition seems increasingly unenforced in many varieties of Indonesian (thus /*gudəg*/ 'jackfruit dish' > [*gudəg*]). The native stratum also shows several further restrictions on vowel co-occurrence: in penultimate syllables, for instance, Lapoliwa (1981) notes that the mid-vowels (his /e o/) never appear when the final syllable hosts /i u/ and that /u/ never appears when the final syllable hosts his /o/. With a handful of exceptions, Mahdi (1981) notes further that the mid-close /e o/ rarely occur in the same words as the mid-open /ɛ ɔ/.

Nasal Substitution

Standard Indonesian has two affixes, the verbal agent voice prefix *meN-* and the nominalizer *peN-*, that trigger a special process on the segments that follow. Before vowel-initial roots, the final segments of these prefixes surface as [ŋ] (thus /*məN-alir*/ 'flow' > [*məŋalir*]), and before voiced obstruents, they assimilate in place (3a)–(3c). Before voiceless stops, however, they assimilate in place and descriptively trigger deletion of the following stop (3d)–(3f), yielding a process termed Nasal Substitution.

(3) *Nasal Substitution*

a. /*məN-bat͡ɕa*/ 'read' > [*mə**mb**at͡ɕa*]	d. /*məN-pɔtɔŋ*/ 'cut' > [*mə**m**ɔtɔŋ*]
b. /*məN-dɔrɔŋ*/ 'push' > [*mə**nd**ɔrɔŋ*]	e. /*məN-tɔlɔŋ*/ 'help' > [*mə**n**ɔlɔŋ*]
c. /*məN-gali*/ 'dig' > [*məŋ**g**ali*]	f. /*məN-kanduŋ*/ 'contain' > [*məŋanduŋ*]

Nasal Substituion operates in very similar ways in Javanese, Jakarta Indonesian (Kurniawan, 2018), and many other languages of Western Indonesia, continuing in all cases an analogous process from Proto-Malayo-Polynesian (Blust, 2004). In Standard Indonesian, it shows two points of variation. Before the segment /s/, first, the final segments of these prefixes typically trigger substitution to /ɲ/ (thus /*məN-sərap*/ 'absorb' > [*məɲərap*]), but in certain loans from English, the process fails and the final nasal assimilates to /n/ (thus /*məN-supsidi*/ 'subsidize' > [*mə**ns**upsidi*]; Sneddon, 1996, pp. 9–12). Before the palatal /t͡ɕ/, in the same vein, the final nasal typically assimilates without triggering substitution (thus /*pəN-t͡ɕut͡ɕi*/ 'washer' > [*pəɲt͡ɕut͡ɕi*]). In both Javanese and Jakarta Indonesian, however, the analogous morphemes do trigger substitution of /t͡ɕ/, yielding forms like [*pəɲut͡ɕi*]). For discussion, see Pater (1996, 1999, 2001, pp. 159–182), Archangeli et al. (1998) and Nomoto (2009).

Nasal substitution shows two further complexities in the wider system of prefixation. First, the process fails to apply between *meN-* and the causative prefix *per-* (thus /*məN-pər-kɔt͡ɕil*/ 'make smaller' > [*mə**mp**ərkɔt͡ɕil*]; the situation is more complex with different varieties of *pər-* and the prefix *peN-* Sneddon, 1996, p. 12). Second, the process does apply between *meN-* and the prefixes *se-* 'one' (thus /*məN-sə-ragam-kan*/ 'standardize' > [*məɲəragamkan*]) and *ke-* (thus /*məN-kə-tawa-i*/ 'laugh at' > [*məŋətawai*]). In Malay, Delilkan (2002) argues that these interactions turn on an asymmetry in the way that strings of prefixes are parsed into metrical feet.

Reduplication

Standard Indonesian shows three types of reduplication. The first and most productive of these is one that copies full prosodic words, yielding patterns like /*pər-əmpat-an*-RED/ 'four-way intersection-RED' > [*pərəmpatan-pərəmpatan*]. Within the verbal system, this process always targets the verbal root and excludes higher voice prefixes, and as a result, it always fails to copy *meN-*, *ber-*, and *di-* (thus /*məN-əlu-RED-kan*/ 'praise repeatedly' > [*məŋəlu-əlukan*]). Even so, it interacts with nasal substitution in a surface-transparent way: when it targets roots prefixed with *meN-* substitution applies in both the base and the reduplicant (thus /*məN-kata-RED-i*/ 'scold repeatedly' > [*məŋ**ata**-ŋ**atai***]; Cohn & McCarthy, 1998; Lapoliwa, 1981). In a second pattern, the language has a small class of roots that show partial reduplication of the string /Cə/: thus *jaka* 'bachelor' ~ *je-jaka* 'bachelor,' *luhur* 'noble' ~ *le-luhur* 'ancestors,' *daun* 'leaf' ~ *de-daunan* 'leaves,' *jamur* 'mushroom' ~ *je-jamuran* 'mushrooms. Third and finally, the language has a productive type of "Rhyming and Chiming Reduplication," analogous to that of Malay (Abdullah, 1974), which copies full prosodic words and

imposes a series of semi-regular phonological chages on either the base or the reduplicant (Lapoliwa, 1981, pp. 189–192). In some cases, these processes target the onset of the second element (*sayur* 'vegetable' > *sayur-**m**ayur* 'all sorts of vegetables,' *lauk* 'side dish' > *lauk-**p**auk* 'all kinds of side dishes'); more often, they target the vowels in the first (*balik* 'return' > *b**o**lak-balik* 'go back and forth'; *ganti* 'change' > *g**o**nt**a**-ganti* 'change back and forth'). For analyses of the phonology beneath this process in Malay, see Soh (2011).

Phonologically Conditioned Allomorphy

The prefixes *ber-*, *ter-*, and *per-* take the forms *be-*, *te-*, and *pe-*before syllables of the shape /cər/ (thus /*bər-kərd͡za*/ 'work' > [*bəkərd͡za*], /*bər-pərgi-an*/ 'go about' > [*bəpərgian*]; /*tər-pərgɔk*/ 'caught' > [*təpərgɔk*]; /*pər-tərnak-an*/ 'animal husbandry' > [*pətərnakan*]). In some cases, the same effect is triggered at a distance by later syllables in the word: thus /*tər-lantar*/ 'abandoned' > [*təlantar*] (Sneddon, 1996, p. 8).

Stress

In many varieties of Indonesian, descriptive sources present a consistent picture of prominence at the level of the word: when morphologically simple words are pronounced in isolation, suspending the influence of higher-level prosodic events, prominence typically falls on the penult and shifts to the ultima when the nucleus of the penult is /ə/. Without question, however, the position of this prominence interacts with a complex phrasal phonology: there are intonational contours that dock on prosodic domains of many different sizes and force prominence to fall on the final syllables of their final words, and within the phonological phrase, these shifts interact with a series of rhythm rules that alter the position of prominence in preceding words as well. In light of these facts, the mainstream view takes the distribution of this prominence to reflect a type of stress that is anchored to the prosodic word but influenced by higher prosodic events (Alieva et al. 1972; Cohn, 1989; Cohn & McCarthy, 1998; Lapoliwa, 1981; Stokhof, 1975; Teeuw, 1984). An alternative strand of work, however, takes these interactions to suggest that the relevant prominence is anchored to a higher prosodic domain, such as the phonological phrase, and thus denies the existence of true word-level stress (Goedemans & van Zanten, 2007; Halim, 1974; van Zanten & van Heuven, 1998), attributing the evidence traditionally marshalled in favor of word-level stress to interference from substrate languges and misperception by native listeners of lnguages that do have word-level stress.

Foot Structure

Beneath the level of the prosodic word, there is clear evidence in Standard Indonesian for word-internal metrical structure and a systematic preference to build disyllabic feet. The case begins most naturally with a type of minimality effect in the system of prefixation: all of the monosyllabic roots in the non-native strata of the lexicon are unable to combine with the prefixes *meN-* and *peN-* in the regular way. Instead, they show the pattern of affixation in (4): beneath *meN-* and *peN-*, the language requires the insertion of the prefix ke- (the same element that- emerges when m*e*N- and the passive *di-* attach to the native root *tau* 'know': thus *di-ke-tau-i* 'known'). The same pattern occurs in many further languages of Western Indonesia, like Javanese (Wedhawati et al., 2001), Sundanese (Robins, 1953), and Riau Indonesian (Gil, 2020), and it is typically interpreted as evidence for a morpheme-specific constraint on foot structure (a type of Prosodic Subcategorization, in the terminology of (Inkelas, 1989): ke- must be introduced before monosyllabic roots because *meN-* and *peN-* require their complements to form disyllabic feet.

(4) *Monosyllabic Roots + meN/peN-: Augmentation*

a. */məN-pɛl/* 'mop' > [*məŋəpɛl*]	d. */məN-bɔr/* 'drill' > [*məŋəbɔr*]
b. */məN-tik/* 'type' > [*məŋətik*]	e. */məN-rɛm/* 'brake' > [*məŋərɛm*]
c. */məN-t͡ɕat/* 'paint' > [*məŋət͡ɕat*]	f. */pəN-bɔm/* 'bomber' > [*pəŋəbɔm*]

In tandem with this effect, it is possible to list at least five further lines of evidence to suggest that Standard Indonesian, like many languages of Western Indonesia, typically requires the right edge of the prosodic word to host a disyllabic foot (and note that there are language games which reveal the existence of disyllabic feet in Jakarta Indonesian, in Riau Indonesian, and in Papuan Malay: (Gil, 2020):

(5) *Arguments for Word-Final Disyllabic Feet:*

a. The vast majority of native roots are disyllabic.
b. Antepenultimate vowels are historically and often synchronically reduced to /ə/, as if there is a process of reduction that is specifically blocked in word-final disyllabic feet.

c. There is a regular process that reduces prevocalic high vowels to glides (/*suatu*/ 'one' > [*swatu*]), and it is specifically blocked whenever it would create monosyllabic words (/*suap*/ 'feed' > [*suwap*], [**swap*]) (Lapoliwa, 1981, pp. 185–187).
d. There is a productive process of blending that typically reduces multi-word strings to two-syllable words, as if its target were this foot (*budak cinta* 'love slave' > *bucin*; *malas gerak* 'too lazy to move' > *mager*; *menteri luar negri* 'foreign minister' > *menlu*).
e. There are intonational contours that spread across a two-syllable window at the right edge of the intonational phrase, like one of the contradiction contours ($bi^{LH}sa^{LH}$ "it is possible").

Prosodic Phrasing

Above the level of words and feet, Halim (1974) argues for the existence of two higher prosodic domains: restricting his attention to South Sumatra Indonesian, he proposes that words are parsed into *pause groups* and *pause groups* are parsed into *intonational patterns*. In the universal prosodic framework of Ito and Mester (2007) and Selkirk (2009), these domains correspond neatly to the categories of *phonological phrase* (ϕ) and *intonational phrase* (ι), and indeed, two very similar prosodic constituents in the related Ambon Malay are equated with the ϕ and ι by Maskikit-Essed and Gussenhoven (2016). In the segmental phonology of Standard Indonesian, many processes seem to be sensitive to these higher domains: In Jakarta Indonesian, for instance, Kurniawan (2018, p. 101) notes that word-final /*h*/ deletes only in words that are non-final in a prosodic domain he identifies as the phrase (in the terms used here, the ι). In Standard Indonesian, in turn, Tadmor (2007, p. 304) alludes to the fact that there are ι intonational tunes that force epenthesis of /ʔ/ after vowel- final words at the right edge of the same domain ι. Beyond this domain, there seem to me to be many further processes that are sensitive to the edge of the ϕ: for instance, the language has a process of coalescence that typically reduces sequences of low and high vowels in hiatus, yielding the reduction of /*mau*/ 'want' to [*mo*] (6a) and /*saja*/ 'me' to [*se*] (6b), and within these words, this process is blocked at the right edge of the ϕ (6c)-(6d). In the same vein, the ϕ and ι seem to impose constraints on the shapes of the feet that fall at their right edges. The right edge of the ϕ, for instance, may well impose a violable requirement that its final foot be disyllabic, as roots like /bəli/ can only undergo ə-deletion outside of ϕ-final positions (7).

(6) *Vowel Coalescence: Conditioned by Position in the Phrase*

a. {ϕ **mo** ini } {ϕ tadi }	c. {ϕ **mau**} {ϕ tadi }
want this earlier	want earlier
'He wanted this earlier.'	'He wanted (it) earlier.'
b. {ϕ **se** suka} {ϕ sih }	d. {ϕ kalo **saja** } {ϕ ndak }
i like actually	as for me not
'I like (it), actually.'	'As for me, I don't.'

(7) *Phrase-Final Position: Stricter Requirements for Disyllabic Feet*

a. {ϕ **bli** ini } {ϕ dulu }	b. {ϕ **bəli** } {ϕ dulu }
buy this first	buy first
'Buy this first!'	'Buy (some) first!'

Conclusion

The central debates in contemporary work on Standard Indonesian lie squarely in the domain of word-level stress, where researchers are beginning to disentangle the audible patterns of low-level prominence from the influence of higher level prosody and substrate phonology. In each of these subdomains, then, it is a fortunate fact that the literature has begun to advance. The study of prosodic phrasing has begun to be taken up in many languages of the region, including especially Waima'a (spoken in East Timor; Himmelmann, 2010) and Mandar (South Sulawesi; Brodkin, To Appear), and fine interactions between foot structure and the edges of the ϕ and ι have begun to emerge in Mandar and other languages of South Sulawesi (Brodkin, 2025a,b). The link between word- and phrase-level prominence, in turn, has been drawn into focus by recent work on Betawi Malay by Van Heuven et al. (2008), Kupang Malay by Lai et al. (2010), Ambon Malay by Maskikit-Essed & Gussenhoven (2016), and Papuan Malay by Kaland (2019, 2021). For valuable review of the central issues in this domain, see Himmelmann &and Kaufman (2018).

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